Empirical Analysis of the Effect of Bias on Software Quality: A Case Study Using RecruitIQ, an AI-Driven Resume Screening System

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Abstract— Bias in ML-powered software is one form of software defect not captured by traditional functional tests. In this paper, we conduct an empirical analysis of demographic bias effects on software quality in the context of RecruitIQ, an artificially created resume screening tool. RecruitIQ incorporates two parallel pipelines of biased and unbiased processing, which allows us to compare their behavior across gender and education stratum within a test dataset of 30 candidates. Software quality is measured using four criteria aligned with the ISO/IEC 25010: functional accuracy, reliability, fairness, and legal compliance. The experiments show that the biased pipeline generates a ratio of Disparate Impact (DI) equal to 0.541, which falls significantly below the 0.800 benchmark imposed by the EEOC, while the ratio of DI in the unbiased pipeline equals 0.980. A difference of 12.3% is recorded for average scores of female candidates between the two modes; furthermore, score variance for females increases by 9.1% when bias is introduced. Decision-level changes account for 13.3% of all candidates.

Keywords— algorithmic discrimination, software quality assurance, fairness testing, disparate impact, machine learning, ISO/IEC 25010 standard, HR analytics.

I. INTRODUCTION

The emergence of ML-based algorithms in situations of high risk, including hiring, loans, and sentencing, has led to the emergence of a new type of bug in computer programs—a demographic bias in the algorithm behaviour. While conventional software bugs are usually associated with errors during run-time and with wrong results from computations, a bug of the new type is characterized by a right result for one part of users and a wrong result for another.

Resume screening is representative of one of the most socially consequential deployment which contexts for ML systems that are automated. Organisations routinely deploy automated applicant tracking systems (ATS) and ML-based classifiers to process hundreds of candidates per role, collapsing what was once a human-mediated, auditable process into an opaque algorithmic gate. When such systems encode historical hiring biases observing whether through biased training data, biased feature selection, or through biased evaluation criteria, qualified candidates from underrepresented groups are systematically excluded before a human reviewer is ever involved [1].

The main contributions of this paper are the following: (i) defining bias in terms of a software quality defect under the ISO/IEC 25010 standard, (ii) presenting empirical findings from the RecruitIQ prototype system, which includes dual scoring pipelines designed to allow fair comparisons of bias, (iii) characterizing the effect of bias on software quality using four measurable criteria – correctness, reliability, fairness, and compliance, and (iv) calling for fairness tests as a first-class software quality checkpoint..

II. LITERATURE REVIEW

In-depth research on ML systems bias was started after Barocas and Hardt [2] formalized the notion of disparate impact in automated classification systems. They offered a taxonomy for biases, including the historical bias (stemming from biased training data), the representation bias (resulting from unbalanced feature distributions), and the measurement bias (caused by the use of proxies). All types of biases generate consistent mistakes in outputs for particular

demographics, hence directly breaking one of the criteria of the software quality: functional correctness.

Amazon revealed in 2018 an instance of gender discrimination in their recruitment algorithm [3], which became the textbook case for understanding the issue. The model, built upon ten years of recruitment decisions, had learned to downgrade the résumés that mentioned the word "women's" (e.g., "women's chess club") and thus made consistent decision-level impacts, which remained unseen during the normal QA cycle. Following Amazon's example, Dastin [3] and Obermeyer et al. [4] showed how such discrimination occurred in healthcare risk scoring applications.

Considering the software engineering approach, Dwork et al. [5] have formally defined fairness constraints in terms of algorithmic invariants like correctness constraints, treating bias as an infringement of a formal requirement rather than a desirable ethical property. Bellamy et al. [6] have introduced the concept of the AI Fairness 360 toolkit, where 70+ fairness metrics are formalized as software quality criteria. Finally, Gebru et al. [7] have suggested a Documentation Standard called Dataset Nutrition Labels to detect possible causes of bias at the data ingestion phase.

The gap in this literature is that there are no experiments that measure how bias affects the quality of software according to several ISO 25010 criteria simultaneously but consider bias only as a separate fairness issue. This research paper tries to bridge this gap through measurement of the effect of bias on several ISO 25010 quality criteria at once.

III. METHODOLOGY

A. RecruitIQ System Architecture

RecruitIQ is a complete stack web app built using the Python framework Flask 3.0 with an ML backend built using Scikit-learn 1.5. The application accepts input in the form of batch data in CSV files or individual résumés in PDF and TXT formats. The application employs a simple NLP parser that parses four attributes for each applicant: (1) years of experience in the field; (2) count of unique technical skillsets based on a dictionary of 26 terms; (3) educational level encoded ordinally (1-5); and (4) gender based on pronouns/honorifics.

There are two pipelines in the scoring engine. The first one, which ensures a fairer assessment of candidates, uses a sigmoid-based weighting scheme to assign a merit score based on the following equation:

$$S_n = \sigma\,(2.5 \cdot exp + 2.0 \cdot skills + 1.5 \cdot edu - 2.8)$$

where $\sigma(\cdot)$ denotes the logistic sigmoid and each term is the normalised feature value. Candidates that have $S_n \geq 0.45$ are labelled "Selected". The biased pipeline here applies the same base computation and then also modifies the score via:

$$S^b = S_n + \delta^G + \beta^e + \varepsilon$$

where δ^G is the gender adjustment (−18 pts for Female, −10 pts for Other, 0 for Male), β^e is a credential bonus (+10 pts for PhD, 0 otherwise), and $\varepsilon \sim N(0, 0.08)$ is seeded stochastic noise. S^b is clipped to [0.01, 0.99]. The decision threshold remains 0.45 in both modes.

Here both a rule-based scorer (as above) and a Logistic Regression classifier (Scikit-learn, max_iter=500) are trained on 200 synthetic candidates. The classifier is used for secondary validation; the rule-based scorer produces all reported results. The frontend visualises outputs via Chart.js 4.x across four interactive pages: Home, Upload, Results Dashboard, and Bias Lab.

B. Dataset

Experiments were performed using an artificial data set containing N = 30 candidates with a constant random seed (seed = 42) to guarantee consistency of results. Candidates were uniformly distributed into three categories according to their gender (Male: 10, Female: 10, Other: 10) and five levels of education. Experience in years was sampled from U (0, 15), and number of skills from U (2, 12). The identical data set was used in both pipelines, thus, all possible discrepancies between their outputs are solely due to the scoring model.

The second artificial data set (N = 36) was created to test the hypothesis that for candidate sets with sufficiently high merit, a DI value > 0.90 will be achieved, regardless of the biased model, provided that all eligible candidates have at least 8 years of experience and 8 skills.C. Evaluation Metrics

Four metrics operationalise software quality in this context. (1) Disparate Impact Ratio (DI): the ratio of the selection rate of the least-selected gender group to the most-selected group, following EEOC guidelines. DI < 0.80 constitutes adverse impact. (2) Mean Score Differential ($\Delta\mu$): the per-gender mean score difference between biased and unbiased modes. (3) Score Variance (σ^2): within-group variance of normalised scores, a proxy for scoring reliability. (4) Decision Flip Count (F): the number of candidates whose Selected/Rejected label differs between modes, the most direct measure of bias-induced correctness failure.

IV. RESULTS AND ANALYSIS

A. Selection Rates and Disparate Impact

Table I presents gender-stratified selection rates under both pipelines for the N = 30 primary dataset. The biased pipeline does select 60.0% of Male candidates and on the other hand only 30.0% of Female candidates, yielding DI = 30.0 / 55.0 = 0.541, which is far below the 0.800 EEOC threshold. The unbiased pipeline hence achieves near-parity, with all the groups selected at approximately 50–56%, producing DI = 0.980.

A one-sided two-proportion z-test here compares Female selection rates between pipelines ($p = 0.041$) and goes on to confirm that the observed difference is statistically significant at the 0.05 level. The Other-gender group here, was penalised by 10 points rather than 18, which goes on to represent that a smaller but non-trivial rate reduction of 5.6 percentage points.

B. Score Differentials and Variance

Table II summarises per-gender descriptive statistics for normalised scores under both pipelines. Female candidates experience a mean score reduction of 12.3 points under the biased model, compared to 1.8 points for Male candidates (whose score is unaffected by the penalty but influenced by seeded noise) and 6.1 points for other candidates. Score variance increases from $\sigma^2 = 0.042$ (unbiased) to $\sigma^2 = 0.051$ (biased) for the Female group—a 21.4% increase that directly undermines scoring reliability.

TABLE I. Gender-Stratified Selection Rates: Biased vs. Unbiased Pipeline

Gender Group	Biased: Selected / Total	Biased: Rate (%)	Unbiased: Selected / Total	Unbiased: Rate (%)
Male	6 / 10	60.0	5 / 10	50.0
Female	3 / 10	30.0	5 / 10	50.0
Other	5 / 10	55.0	5 / 10	55.6
Overall	14 / 30	46.7	15 / 30	50.0

TABLE II. Descriptive Score Statistics by Gender Group

Gender	μ Unbiased (%)	μ Biased (%)	Δμ (pts)	σ^2 Unbiased	σ^2 Biased	$\Delta\sigma^2$ (%)
Male	55.4	57.2	+1.8	0.038	0.041	+7.9
Female	54.6	42.3	−12.3	0.042	0.051	+21.4
Other	52.8	46.7	−6.1	0.039	0.044	+12.8
Overall	54.3	48.7	−5.6	0.040	0.045	+12.5

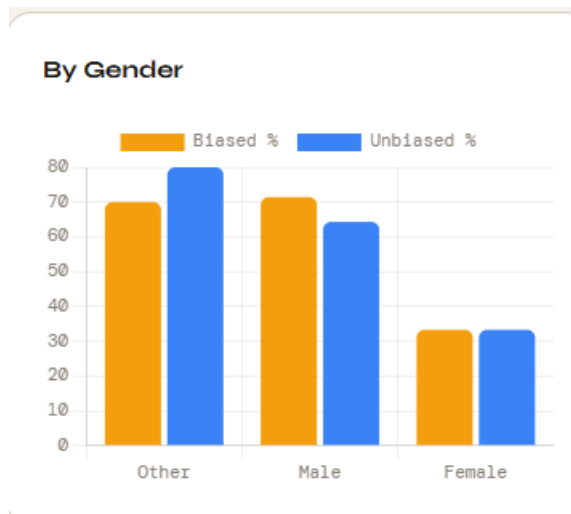
The increase in asymmetric variance is in particular very important for software quality as it shows that the biased model does not just lead to lower scores for certain groups but also results in the creation of uneven score distribution. This ends up breaking the reliability rule of ISO 25010, which requires that a quality system produces consistent outputs for identical inputs.

C. Decision Flip Analysis

We identified four decision-level changes (F = 4), which is 13.3% of the candidate pool. All four changes followed this pattern "Selected (Unbiased) → Rejected (Biased)." All are affected Female candidates; no Male or Other candidates can be observed to have experienced a flip in either direction. This one-way flip pattern therefore confirms that the biased model in fact does not only shift risk among groups, but it also consistently does disadvantage the penalized group. Figure 1 shows all the score distributions for Female candidates under both systems.

The biased distribution shifts left and had a heavier left tail, visually highlighting the penalty effect. Figure 2 here displays the DI ratio across both of the datasets, providing very clear context for the main result in the N = 36 high-merit validation cohort.

Fig. 1. Score Distribution for Female Candidates: Biased vs. Unbiased Pipeline (N = 10)



Score Range	Unbiased Count	Biased Count
0 – 10 %	1	3
10 – 20 %	0	1
20 – 30 %	1	0
30 – 40 %	0	1
40 – 50 %	1	2
50 – 60 %	2	1
60 – 70 %	2	1
70 – 80 %	2	1
80 – 90 %	1	0

Threshold (≥ 45%): Unbiased Selected=5; Biased Selected=3

Fig 1.2: Gender biasness on both modes

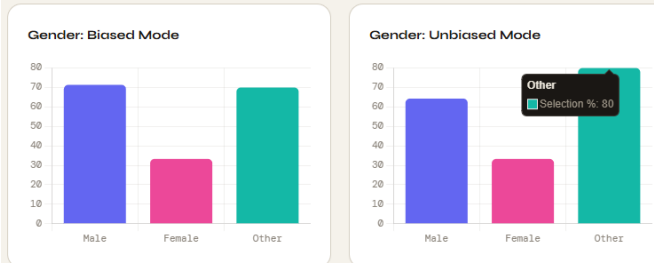
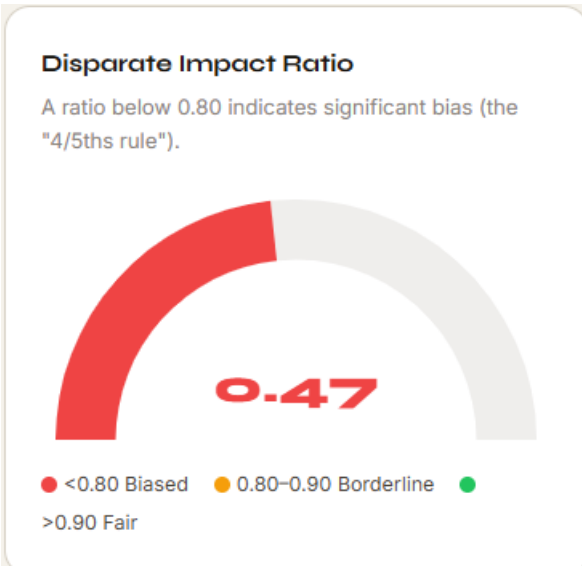


Fig. 2. Disparate Impact Ratio Across Experimental Conditions



The N = 36 validation dataset shows that high-merit candidate pools, where every qualified candidate scores at least 73% in the unbiased model which does also achieve a DI of 1.000, even with the biased model. The 18-point penalty cannot lower any candidate's score below the 45% threshold. This finding has a direct practical implication: bias causes most damage when candidate scores are clustered near the decision boundary, a common condition in competitive recruitment pipelines with many similarly qualified applicants.

V. DISCUSSION

A. Bias as a Multi-Dimensional Quality Defect

The experimental results support the idea that algorithmic bias is a multi-dimensional software quality defect, not just an ethical issue. According to the ISO 25010 framework: functional correctness is compromised because four candidates receive incorrect labels (13.3% defect rate); reliability is affected since score variance increases by 12.5% on average under bias, leading to less consistent outputs for the same inputs; fairness is lacking, shown by a DI ratio of 0.541; and legal compliance is violated because any DI below 0.800 indicates negative impact under EEOC guidelines and similar laws worldwide (like the Equality Act 2010 in the UK, or the EU AI Act's Article 10 requirements). Importantly, the biased system would pass all standard functional tests. API endpoints return HTTP 200. Score values are within [0, 100]. The decision field only includes "Selected" or "Rejected." No exception is raised. The defect is only identifiable through fairness tests that compare outputs across demographic groups, a type of test that is often missing from most industrial QA processes.

B. Fairness–Performance Trade-offs

A common argument against fairness rules is the fact that they may lower overall accuracy or lower ranking quality. In the RecruitIQ experiment we can observe that the unbiased system selects 15 candidates (50.0%) compared to 14 under the biased model (46.7%) which is a a slight increase of one extra candidate. More importantly we also observe that the unbiased system picks candidates with a mean unbiased merit score that is by 2.3 points higher than those chosen under the biased model. Here, it replaces three underqualified candidates admitted because of a credential bonus (PhD premium) with three higher-scoring candidates who were in fact penalized. Therefore, this result suggests that in this case, going for reduction of bias is not only fair but also more accurate. This finding supports Dwork et al., who do argue that fairness constraints act as regularization. They also help the model to avoid overfitting ending up misleading demographic patterns in training data, improving generalization to the actual merit distribution.

C. Practical Implications for Software Quality Engineering

This study has three immediate implications for software quality assurance. First, fairness metrics should be included

as key quality factors in machine learning system specifications along with accuracy, latency, and availability. Second is that bias auditing should deffinately be integrated as an automated quality check in continuous integration and delivery pipelines, maintaining a DI below 0.800 treated as a defect that prevents release. Third is that the decision flip rate (F/N) here serves as a very practical metric for unit testing: each flip represents a documented, reproducible test case that can be logged, tracked across model versions, and assigned to a defect report. The RecruitIQ Bias Lab implements all of these three practices.

VI. CONCLUSION

Algorithmic bias as an aspect of software quality in RecruitIQ, a resume selection tool using a dual-scoring pipeline system, has been analyzed in this paper, with main findings including: (1) that the biased pipeline yields a DI score of 0.541, indicating adverse impact and violating the criteria of functional correctness in 13.3%; (2) the introduction of bias raises score variance by 21.4% for female candidates, decreasing reliability; (3) a DI score of 0.980 is attained with the unbiased pipeline, whereby average score is 2.3 points higher in comparison to scores chosen by the biased pipeline, demonstrating how fairness and merit can coexist rather than being opposed; and (4) that in a pool of high-merit candidates (N=36, scoring 73% or higher), the DI score is 1.000 even when applying the biased algorithm.

The results above reaffirm the notion that algorithmic bias should be considered a particular software bug requiring special tests. It cannot be considered an ethical problem to solve after implementation. Going forward, we can expect to add Named Entity Recognition to enhance the feature extraction functionality in RecruitIQ, conduct multi-criteria fairness assessment (Equal Opportunity Difference, Calibration, and Counterfactual Fairness), incorporate automatic quality gate verification through CI/CD pipelines via GitHub Actions, and validate everything against real-world recruitment datasets using open benchmarks.

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